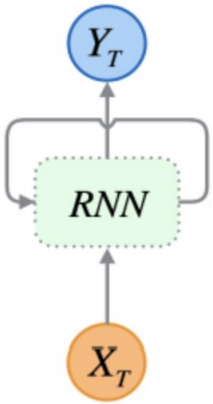
CIS 678 Machine Learning

Special Topics

Recurrent Neural Networks

Originally designed to Solve:

- Sequential Problems
  - *NLP (MT, ...), Speech Recognition,...*
  - *DNA Sequencing*
- Time-series Problems



ref

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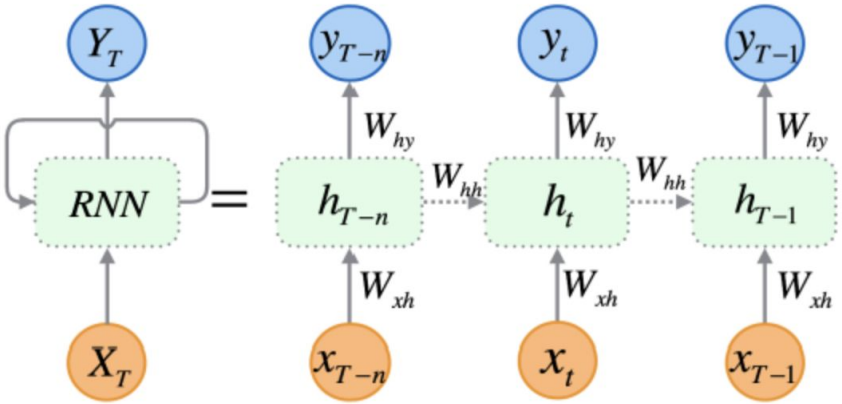

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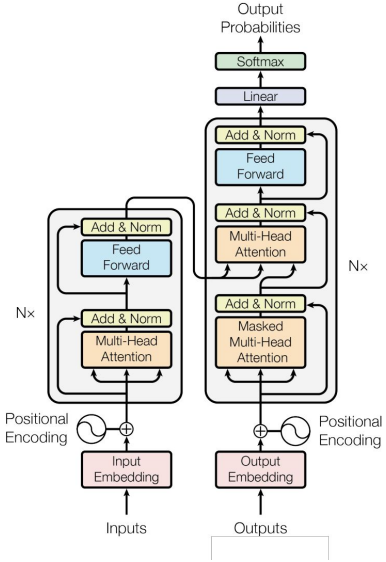
Transformers

Examples:

- *Encoder decoder pair*
- *GPT*
- *BERT*

BERT

Encoder



GPT

Decoder

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Generative AI

Examples:

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)

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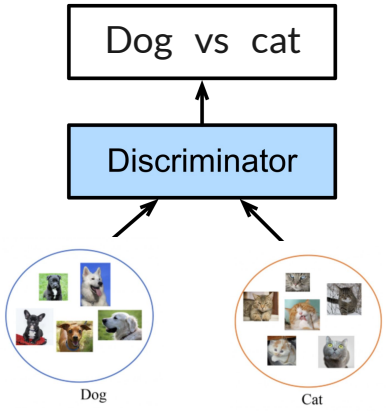
Generative AI



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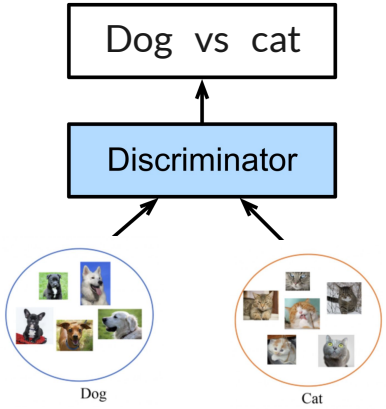
Generative AI

Examples:

- Generative Adversarial Networks (GANs)



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Generative AI

Examples:

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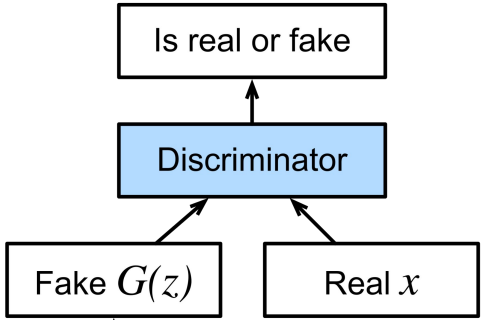

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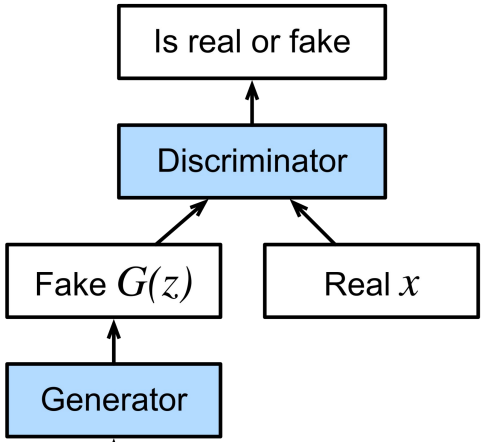

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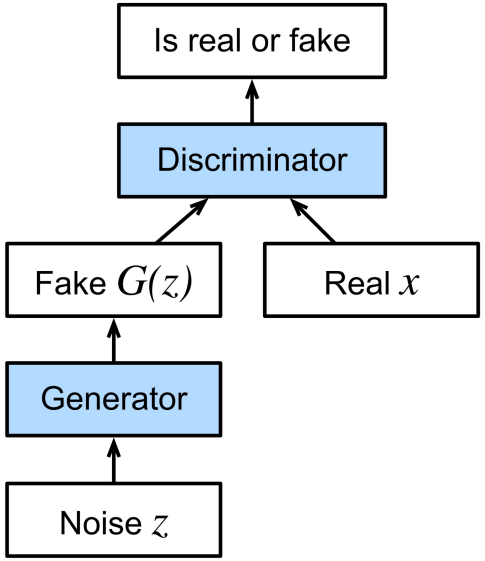

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Generative AI

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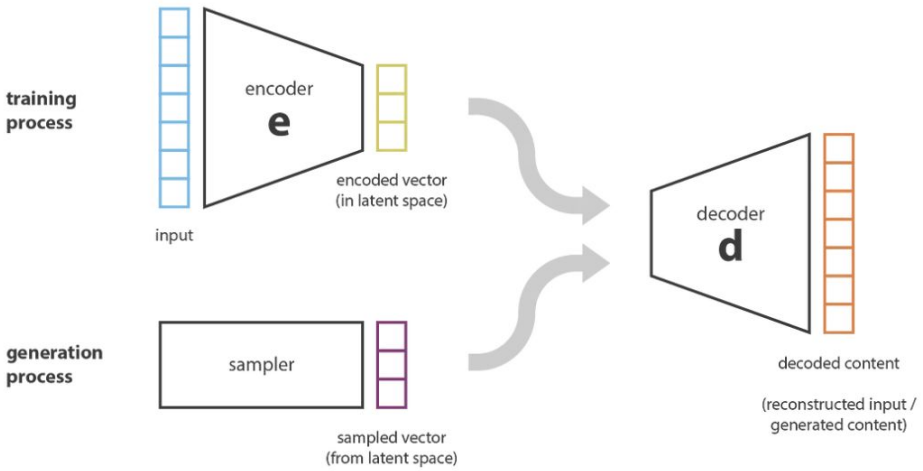

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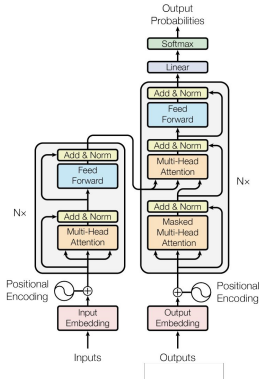
Foundational models

- We don’t train independent models explicitly (say machine translation)

BERT

Encoder



Machine translation

GPT

Decoder

Foundational models

- We don’t train independent models explicitly (say machine translation)
- Or document summarization

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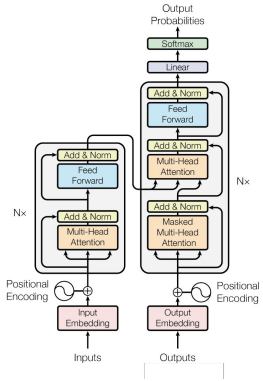

GPT

Decoder

Document summarization

Foundational models

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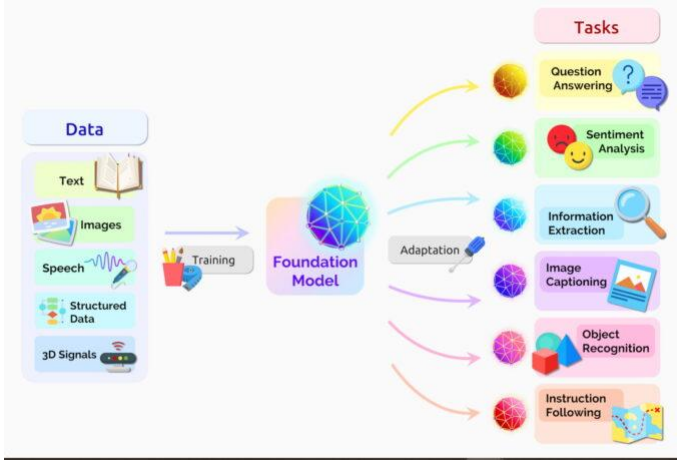
- **We train a Base model for multiple tasks jointly, and then fine tune for specific tasks**



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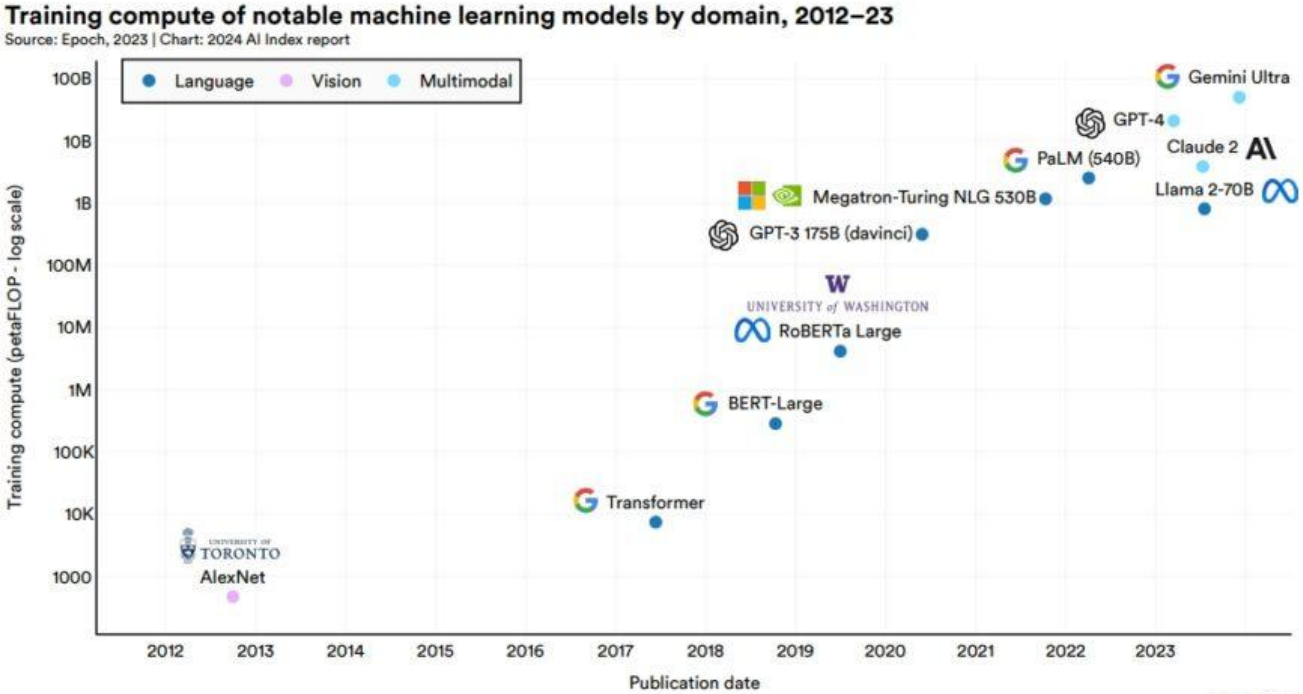

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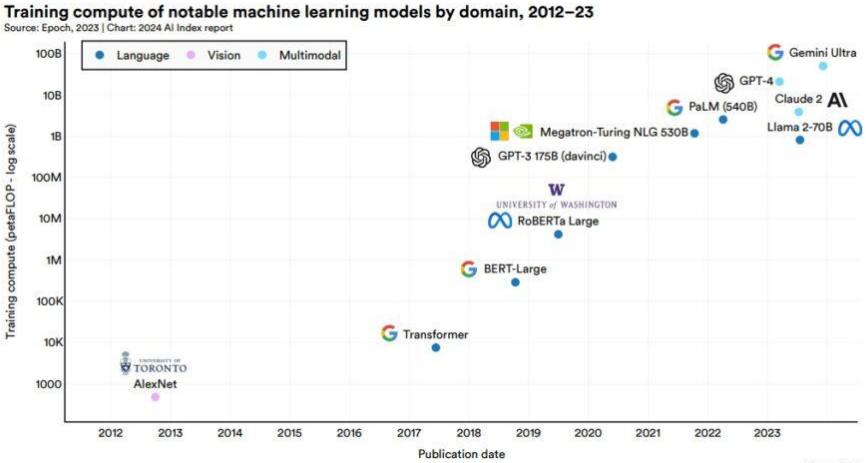

Figure 1.3.7

Training Neural Network Challenges

- Intractable gradients
  - Vanishing, and
  - Exploding gradients

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- Various normalizations
  - Input normalization (standard scalar)
  - Batch normalization
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- Intractable gradients
  - Vanishing, and
  - Exploding gradients
- Various normalizations
  - Input normalization (standard scalar)
  - Batch normalization
  - Layer normalization
- Controlling overfitting
  - Regularization
  - Early stopping
  - Drop out

QA